Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/08/2016
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	11/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			
	Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany		

Task Description:	Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• Visiting scientist (March-June 2016) at DLR-Koln Germany to conduct undercooling experiments and metallography of quenched samples to support flight investigations for Batch 1 (FeCo, FeCrNi), 2 (FeCo, TiZrNi, NdFeB), Batch 2 (FeB, FeC), and Batch 3 (FeCrNi, CoSi, NiSn) samples. • ISS increment 47 PI (FeCrNi), MUSC Control Center DLR Köln DE, 4/25-5/02 (2016) conducted flight space tests on MSL-EML to define viscosity as a function of temperature to support MHD modeling activity and investigated the influence of convection on a ternary eutectic alloy. • ISS increment 49 co-I (MC2), MUSC Control Center DLR Köln DE, 7-9 November, conducted MSL-EML flight space testing with Rainer Wunderlich of Ulm to compare to ground-based results obtained at NASA/MSFC (Marshall Space Flight Center) ESL. • Conducted ground-based electrostatic levitation (ESL) flight support testing at MSFC Feb 2016. • Conducted ground-based electrostatic levitation (ESL) flight support testing at MSFC May 2016. • Conducted ground support testing to monitor EML telemetry from ESA during real-time ISS operations using an eToken from DLR/MUSC. Developed software to digitize the analog signal from the DLR control room to a dedicated computer at Tufts University for analysis on May 19 (2016). • Conducted DESY Synchrotron campaign team tests (FeCrNi/CoSi), Hamburg Germany, 15-18 April, to define phase transformation sequence and incubation time. • Conducted ground-based EML density measurement using DLR EML with Jürgen Brillo and Yuriy Plevachuk of Ivan Franko National University, Lviv, Ukraine. • Conducted ground-based non-levitation viscometry testing of steel alloy to provide baseline results for comparison to space testing with Jürgen Brillo of DLR. • Visited Mikhail Krivilev at Udmurt State University in Izhevsk Russia 1-7 July to support paper on nucleation kinetics with stirring. • TMS 2016 Annual meeting – Materials Science in Reduced Gravity Symposium, Nashville TN, Feb 17-18, 2016, Symposium Organizer. • Attended ORNL Beamline NESL workshop to investigate possible use of levitation diffractometry for phase transformation kinetic path tracking. • Attended Investigators Working Group (IWG)#14 24-25 May and IWG#15 12-13 December at DLR Köln. • Attended ASGSR (American Society for Gravitational and Space Research) Annual conference as symposium organizer and presenter.
Bibliography Type:	Description: (Last Updated: 05/05/2026)

Articles in Peer-reviewed Journals	Matson DM, Watanabe M, Pottlacher G, Lee GW, Fecht HJ. "Thermophysical property measurement: A call to action." International Journal of Microgravity Science and Application. 2016 Sep;33(3):330304-1 to 7. , Sep-2016
Articles in Peer-reviewed Journals	Matson DM, Xiao X, Rodriguez J, Wunderlich RK. "Preliminary experiments using electromagnetic levitation on the International Space Station." International Journal of Microgravity Science and Application. 2016 Sep;33(2):330206-1 to 11. http://dx.doi.org/10.15011/jasma.33.330206 , Sep-2016
Articles in Peer-reviewed Journals	Sansoucie MP, Rogers JP, Kumar V, Rodriguez J, Xiao X, Matson DM. "Effects of environmental oxygen content and dissolved oxygen on the surface tension and viscosity of nickel." International Journal of Thermophysics. 2016 Jul;37:76,1-11. http://dx.doi.org/10.1007/s10765-016-2085-6 , Jul-2016
Dissertations and Theses	Rodriguez J. "Thermodynamic and Growth Kinetic Modeling of Solidification Form a Metastable Mushy Zone." PhD Thesis, Tufts University Mechanical Engineering, Medford, MA, August 2016. , Aug-2016
Papers from Meeting Proceedings	Sansoucie MP, Rogers JP, Matson DM. "Rapid Quench in an Electrostatic Levitator." TMS 2016. 145th Annual Meeting, The Minerals, Metals and Materials Society, Nashville, TN, February 14-18, 2016. TMS 2016 Supplemental Proceedings. Hoboken, NJ : Wiley for The Minerals, Metals and Materials Society, 2016. p. 31-34. , Mar-2016